

System Architecture & RAG Pipeline Technical Manual

Author: Cloud Systems Architecture Group | Version: 2.1 | Module: High-Performance Vector Retrieval

1. System Overview & Tech Stack

The platform is engineered as a microservice-based RAG infrastructure utilizing Python 3.11+, FastAPI web framework, LangChain 0.2+, and Supabase Vector (pgvector extension). The architecture prioritizes low latency (<250ms vector query turnaround), high precision retrieval (MRR > 0.85), and real-time streaming answers.

2. Vector Embedding Engine Benchmarks

The vector store supports two embedding options configured dynamically per request:

- HuggingFace sentence-transformers/all-MiniLM-L6-v2: Outputs 384-dimensional dense vectors. Runs locally on CPU/GPU with zero external dependency. Best for high-throughput, zero-cost internal indexing.
- OpenAI text-embedding-3-small: Outputs 1536-dimensional dense vectors. Requires valid OPENAI_API_KEY. Delivers superior multi-lingual semantic retrieval for technical manuals and legal contracts.

3. Vector Database Architecture & Indexing

3.1 Local Storage (ChromaDB): Default local engine persists embeddings on disk in SQLite + HNSW graph structure. Collection distance metric is set to Cosine Similarity ($1 - \text{cosine distance}$).

3.2 Cloud Production (Supabase pgvector): Enterprise instances connect via PostgreSQL driver with IVFFlat or HNSW vector index algorithms enabling sub-10ms similarity searches across 10,000,000+ vector chunks.

4. RAG Query Pipeline & Citation Formatting

When a user query arrives at POST /api/chat:

1. The query text is sanitized and passed to the embedding engine to compute `query_vector`.
2. The Vector Database executes a top-K similarity search (default K=4) filtering out chunks with similarity score below threshold (default score threshold = 0.35).
3. Top chunks are formatted into a context block with explicit markers: [Source: filename, Page: X, Chunk ID: Y].
4. The augmented system prompt instructs the LLM: 'Answer strictly using the provided context. If the answer cannot be deduced from the context, state clearly that the information is unavailable.'
5. The streaming response generator streams text tokens while appending an explicit citations array containing source filename, page number, confidence match percentage, and matched text snippet.

5. Troubleshooting & Maintenance

If retrieval precision degrades, run vector re-indexing with chunk size 750 and overlap 100. Check logs at `/var/log/rag_service.log` for embedding API rate limit errors.